

CPSC 332 Project

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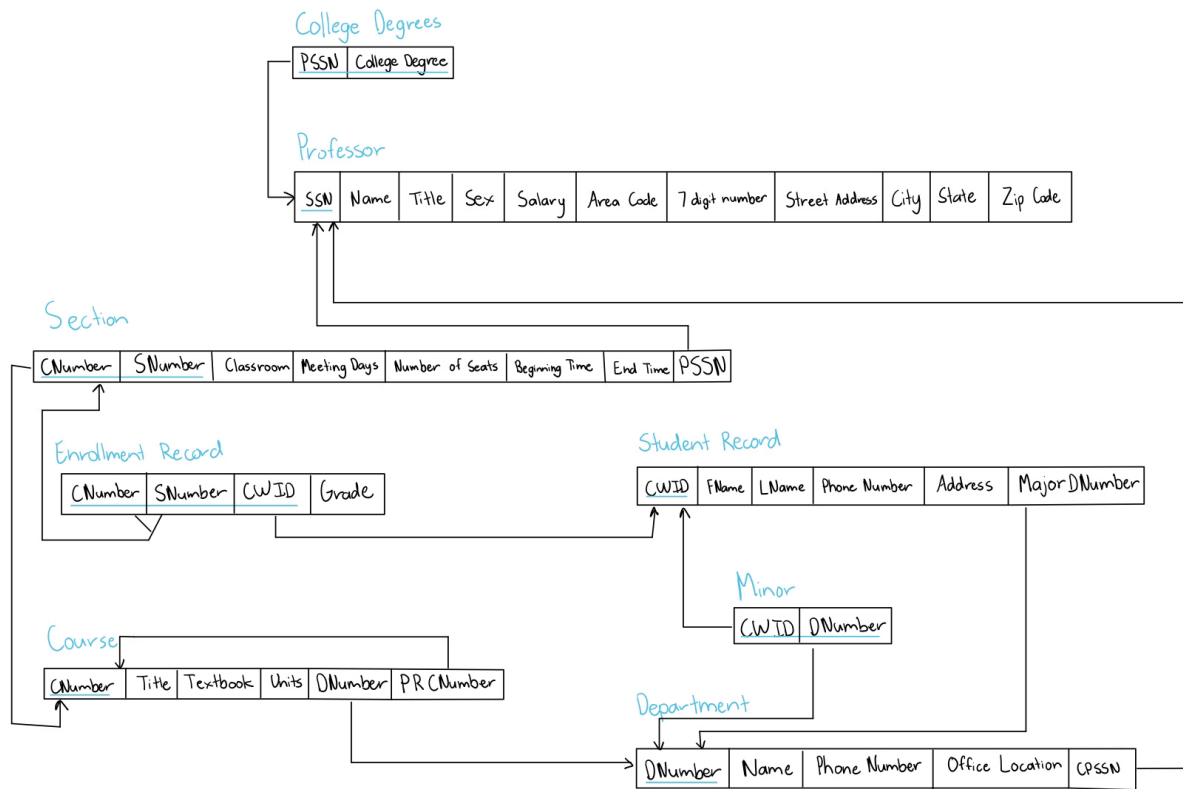
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Account with Project Implementation:

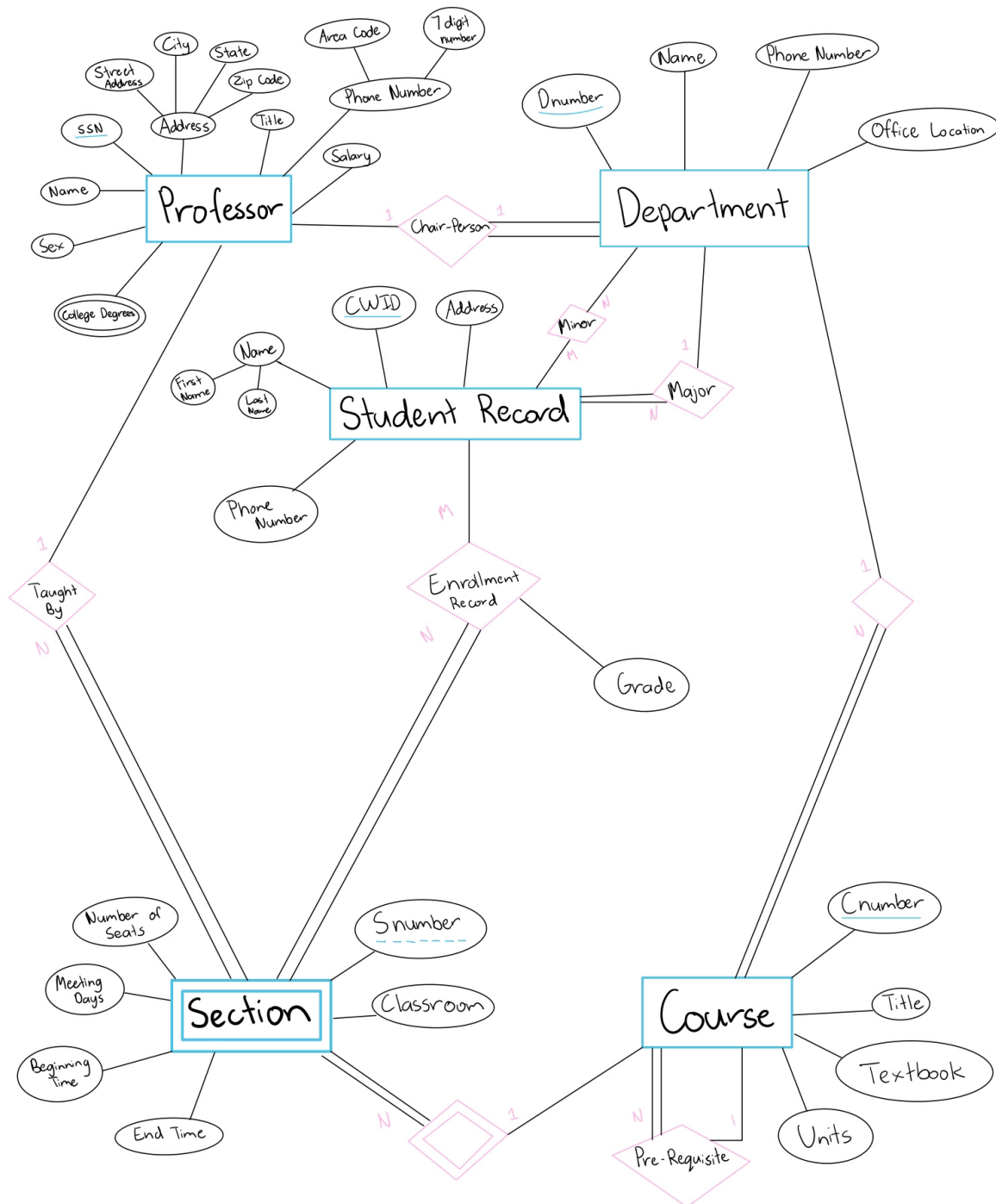
cs332h11

<https://ecs.fullerton.edu/~cs332h11/>

ER Diagram



Relational Model



DDL

```
DROP TABLE IF EXISTS student_record;  
DROP TABLE IF EXISTS minor;  
DROP TABLE IF EXISTS professor;  
DROP TABLE IF EXISTS prof_degree;  
DROP TABLE IF EXISTS course;  
DROP TABLE IF EXISTS section;  
DROP TABLE IF EXISTS enrollment;  
DROP TABLE IF EXISTS department;  
DROP TABLE IF EXISTS prereqCourse;
```

```
CREATE TABLE student_record (  
    cwid VARCHAR(255),  
    fname VARCHAR(255),  
    lname VARCHAR(255),  
    phone VARCHAR(255),  
    addy VARCHAR(255),  
    major_dno VARCHAR(255)  
);
```

```
CREATE TABLE minor (  
    cwid VARCHAR(255),  
    dnum VARCHAR(255)  
);
```

```
CREATE TABLE professor (  
    ssn VARCHAR(255),  
    prof_name VARCHAR(255),  
    title VARCHAR(255),  
    sex VARCHAR(255),  
    salary VARCHAR(255),  
    area_code VARCHAR(255),  
    phone_num VARCHAR(255),  
    street_addy VARCHAR(255),  
    city VARCHAR(255),  
    prof_state VARCHAR(255),  
    zip VARCHAR(255)
```

```
);  
CREATE TABLE prof_degree (  
    Prof_ssn VARCHAR(255),  
    degree VARCHAR(255)  
);  
  
CREATE TABLE course (  
    cnum VARCHAR(255),  
    title VARCHAR(255),  
    textbook VARCHAR(255),  
    units VARCHAR(255),  
    dnum VARCHAR(255),  
    preqCnum VARCHAR(255)  
);  
  
CREATE TABLE prereqCourse (  
    cnum VARCHAR(255),  
    prereq VARCHAR(255)  
);  
  
CREATE TABLE section (  
    cnum VARCHAR(255),  
    snum VARCHAR(255),  
    classroom VARCHAR(255),  
    meeting_days VARCHAR(255),  
    num_of_seat VARCHAR(255),  
    begin_time VARCHAR(255),  
    end_time VARCHAR(255),  
    prof_ssn VARCHAR(255)  
);  
  
CREATE TABLE enrollment (  
    cnum VARCHAR(255),  
    snum VARCHAR(255),  
    cwid VARCHAR(255),  
    grade VARCHAR(255)  
);
```

```

CREATE TABLE department (
    dnum VARCHAR(255),
    dname VARCHAR(255),
    phone VARCHAR(255),
    office_location VARCHAR(255),
    chair_ssn VARCHAR(255)
);

--student tables--

--INSERT INTO student_record VALUES
('cwid','fname','lname','phone','addy','majordno');
INSERT INTO student_record VALUES
('1231','powder','nguyen','123-456-7890','123 Sesame Street, Unicorn
Valley','2244');
INSERT INTO student_record VALUES
('1232','jinx','labinagin','234-567-8910','234 Sesame Street,
Unicorn Valley','2244');
INSERT INTO student_record VALUES
('1233','vi','lim','345-678-9012','345 Sesame Street, Unicorn
Valley','2244');
INSERT INTO student_record VALUES
('1234','caitlyn','pham','456-789-0123','456 Sesame Street, Unicorn
Valley','2244');
INSERT INTO student_record VALUES
('1235','ekko','dang','567-890-1234','567 Sesame Street, Unicorn
Valley','3366');
INSERT INTO student_record VALUES
('1236','jayce','tran','678-901-2345','678 Sesame Street, Unicorn
Valley','3366');
INSERT INTO student_record VALUES
('1237','vander','smith','789-012-3456','789 Sesame Street, Unicorn
Valley','3366');
INSERT INTO student_record VALUES
('1238','isha','washington','012-345-6789','890 Sesame Street,
Unicorn Valley','3366');

--department tables--

```

```

--INSERT INTO department VALUES
('dnum','dname','phone','location','chairssn');
INSERT INTO department VALUES
('2244','compsci','951-123-4567','ecs','2341');
INSERT INTO department VALUES
('3366','math','401-103-0203','mccarthy','2342');

--professor tables--
--INSERT INTO professor VALUES ('ssn', 'prof_name', 'title', 'sex',
'salary', 'area_code', 'phone_num', 'street_addy', 'city',
'prof_state', 'zip' )
INSERT INTO professor VALUES
('2341','Heimerdinger','Dr','male','100k','951','800-567-9832','123
Mimi Way','Fullerton','CA','92913');
INSERT INTO professor VALUES
('2342','Cheawon','Dr','female','120k','123','124','124 Lala Land',
'Happy City','CA','92640');
INSERT INTO professor VALUES
('2343','Hanni','Dr','female','300k','951','125','125 Disney Land',
'Temecula','CA','92592');

--course tables--
--INSERT INTO course VALUES
('cnum','title','textbook','unit','dnum','preqcnun');

INSERT INTO course VALUES ('332','database','database
textbook','3','2244','none');
INSERT INTO course VALUES ('335','algo','algo
textbook','3','2244','none');
INSERT INTO course VALUES ('338','stats','stats
textbook','4','3366','none');
INSERT INTO course VALUES ('150','calc','calc
textbook','4','3366','none');

--sections--
--INSERT INTO section VALUES ('cnum','snum','classroom','meeting
days','seats','begintime','endtime','prof');

```

```

INSERT INTO section VALUES
('332','1','39','Sunday','30','10am','12pm','2341');
INSERT INTO section VALUES
('335','2','40','Monday','35','10am','12pm','2341');
INSERT INTO section VALUES
('338','3','41','Tuesday','35','10am','12pm','2341');
INSERT INTO section VALUES
('150','4','42','Wednesday','35','10am','12pm','2342');
INSERT INTO section VALUES
('332','5','43','Thursday','35','10am','12pm','2342');
INSERT INTO section VALUES
('150','6','44','Friday','30','10am','12pm','2343');

--enrollment record--

--INSERT INTO enrollment VALUES ('cnum','snum','cwid','grade');
INSERT INTO enrollment VALUES ('332','1','1231','A');
INSERT INTO enrollment VALUES ('332','1','1232','A');
INSERT INTO enrollment VALUES ('332','1','1234','A');
INSERT INTO enrollment VALUES ('332','1','1233','C-');
INSERT INTO enrollment VALUES ('332','1','1235','C+');

INSERT INTO enrollment VALUES ('335','2','1231','A');
INSERT INTO enrollment VALUES ('335','2','1232','A');
INSERT INTO enrollment VALUES ('335','2','1236','A+');
INSERT INTO enrollment VALUES ('335','2','1237','B-');
INSERT INTO enrollment VALUES ('335','2','1238','B');

INSERT INTO enrollment VALUES ('338','3','1231','C');
INSERT INTO enrollment VALUES ('338','3','1232','C');
INSERT INTO enrollment VALUES ('338','3','1233','D');
INSERT INTO enrollment VALUES ('338','3','1234','D');
INSERT INTO enrollment VALUES ('338','3','1235','A+');

INSERT INTO enrollment VALUES ('332','5','1237','A');
INSERT INTO enrollment VALUES ('332','5','1238','A');
INSERT INTO enrollment VALUES ('150','6','1237','B');
INSERT INTO enrollment VALUES ('150','6','1238','B-');
INSERT INTO enrollment VALUES ('150','6','1232','B-');

```


SQL Queries

```
--test for query a of student--
```

```
SELECT section.cnum, section.classroom, section.meeting_days,  
section.begin_time, section.end_time, COUNT(*) AS student_count  
FROM section, enrollment  
WHERE section.cnum = '$input' AND enrollment.snum = section.snum  
GROUP BY section.cnum, section.classroom, section.meeting_days,  
section.begin_time, section.end_time;  
--test for query b of student--
```

```
SELECT course.title, enrollment.grade  
FROM student_record, course, enrollment  
WHERE student_record.cwid = '$input' and student_record.cwid =  
enrollment.cwid and enrollment.cnum = course.cnum;
```

```
--test query for a of prof--
```

```
SELECT course.title, section.classroom, section.meeting_days,  
section.begin_time, section.end_time  
FROM course, section  
WHERE section.prof_ssn = '$input' and section.cnum=course.cnum;
```

```
--test query for b of prof--
```

```
SELECT enrollment.grade, COUNT(*) as studnets  
FROM enrollment  
WHERE enrollment.cnum = '$input1' AND enrollment.snum = '$input2'  
GROUP BY enrollment.grade;
```

sql_connection.php

```
<?php
// Gets a connection to the database
const username = '';
const password = '';
const hostname = '';
const database_name = '';

$link = mysqli_connect(hostname, username, password, database_name);
if (!$link) {
    die("Unable to connect to database: " . mysqli_connect_error());
}
echo 'Connected successfully<p>';
?>
```

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Document</title>
</head>
<header>
    <h1>332 Database Project</h1>
</header>

<nav>
    <a href="professor.html">Professor Portal</a>
    <a href="student.html">Student Portal</a>
</nav>
<body>

</body>
</html>
```

student.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>Student Homepage</title>
</head>
<body>
  <header>
    <h1>Student Portal</h1>
    <h2>Find Course Sections</h2>

    <form action="student.php" method="post">
      <label for="cnum">Course Number:</label>
      <input type="text" name="cnum">
      <button type="submit">Search</button>
    </form>

    <h2>Find Grades</h2>

    <form action="student.php" method="post">
      <label for="cwid">Campus-Wide ID:</label>
      <input type="text" name="cwid">
      <button type="submit">Search</button>
    </form>

  </header>

  <nav>
    <a href="index.html">Back to Home</a>
  </nav>
</body>
</html>
```

student.php

```
<?php
// Gets connection to database
require_once('sql_connection.php');
// If cnum is set run student A query
if (isset($_POST['cnum'])) {

    $input = $_POST['cnum'];
    $query = "SELECT section.cnum,
section.snum,section.classroom, section.meeting_days,
section.begin_time, section.end_time, COUNT(*) AS student_count
            FROM section, enrollment
            WHERE section.cnum = '$input' AND enrollment.snum
= section.snum
            GROUP BY section.snum, section.classroom,
section.meeting_days, section.begin_time, section.end_time";

    $result = $link->query($query);

    if ($result->num_rows > 0) {
        // Loop through each row in the result set
        while ($row = $result->fetch_assoc()) {
            // Display
            echo "Section Number: " . $row["snum"] . " -
Classroom: " . $row["classroom"] . " - Meeting Days: " .
            $row["meeting_days"] . " -
Time: " . $row["begin_time"] . "-" .
            $row["end_time"] . " -
Enrolled: " . $row["student_count"] . "<br>";
        }
    } else {
        echo "No results found.";
    }
}

// If cwid is set run student B query
if (isset($_POST['cwid'])) {
```

```

$input = $_POST['cwid'];
$query = "SELECT course.title, enrollment.grade
          FROM student_record, course, enrollment
          WHERE student_record.cwid = '$input' and
student_record.cwid = enrollment.cwid and enrollment.cnum =
course.cnum";

$result = $link->query($query);

if ($result->num_rows > 0) {
    // Loop through each row in the result set
    echo "Courses Taken:<br>";
    while ($row = $result->fetch_assoc()) {
        // Display
        echo "Course: " . $row["title"] . " - Grade: " .
$row["grade"] . "<br>";
    }
} else {
    echo "No results found.";
}
}
$result->free_result();
$link->close();
?>

<html>
  <body>

    <nav>
      <a href="index.html">Back to Home</a>
    </nav>

  </body>
</html>

```

professor.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>Professor Homepage</title>
</head>
<body>
  <header>
    <h1>Professor Portal</h1>
    <h2>List Classes</h2>

    <form action="professor.php" method="post">
      <label for="ssn">SSN:</label>
      <input type="text" name="ssn">
      <button type="submit">Find Classes</button>
    </form>

    <h2>List Student Grades</h2>
    <form action="professor.php" method="post">
      <label for="cnum">Course Number:</label>
      <input type="text" name="cnum"> <br>
      <label for="snum">Section Number:</label>
      <input type="text" name="snum">
      <button type="submit">Submit</button>
    </form>

  </header>

  <nav>
    <a href="index.html">Back to Home</a>
  </nav>
</body>
</html>
```

professor.php

```
<?php
// Get connection
require_once('sql_connection.php');
// If ssn is set, run the query a of professor
if (isset($_POST['ssn'])) {
    $input = $_POST['ssn'];

    $query = "SELECT course.title, section.classroom,
section.meeting_days, section.begin_time, section.end_time
              FROM course, section
              WHERE section.prof_ssn = '$input' and
section.cnum=course.cnum;";

    $result = $link->query($query);

    if ($result->num_rows > 0) {
        // Loop through each row in the result set
        while ($row = $result->fetch_assoc()) {
            // Display
            echo "Course: " . $row["title"] . " - Classroom: " .
$row["classroom"] .
            " - Days: " . $row["meeting_days"] . " - Time: "
.$row["begin_time"] . "-" . $row["end_time"] . "<br>";
        }
    } else {
        echo "No results found.";
    }
}

// if cnum is set run query b of professor
if (isset($_POST['cnum'])) {
    // since cnum is set we assume that snum will also be set
    $input = $_POST['cnum'];
    $input2 = $_POST['snum'];

    $query = "SELECT enrollment.grade, COUNT(*) as studnets
              FROM enrollment
              WHERE enrollment.cnum = '$input' AND
```

```

enrollment.snum = '$input2'
        GROUP BY enrollment.grade";

$result = $link->query($query);

if ($result->num_rows > 0) {
    // Loop through each row in the result set
    while ($row = $result->fetch_assoc()) {
        // Display
        echo "Grade (" . $row["grade"] . "): " . "Count: "
.$row["studnets"] . "<br>";
    }
} else {
    echo "No results found.";
}
}
$result->free_result();
$link->close();
?>

<html>
    <body>

    <nav>
        <a href="index.html">Back to Home</a>
    </nav>

    </body>
</html>

```


332 Database Project

[Professor Portal](#) [Student Portal](#)

Professor Portal

List Classes

SSN:

List Student Grades

Course Number:
Section Number:

[Back to Home](#)

Student Portal

Find Course Sections

Course Number:

Search

Find Grades

Campus-Wide ID:

Search

[Back to Home](#)

Professor Sample Input/Output

Query A:

Professor Portal

List Classes

SSN:

List Student Grades

Course Number:

Section Number:

[Back to Home](#)

Connected successfully

Course: database - Classroom: 39 - Days: Sunday - Time: 10am-12pm

Course: algo - Classroom: 40 - Days: Monday - Time: 10am-12pm

Course: stats - Classroom: 41 - Days: Tuesday - Time: 10am-12pm

[Back to Home](#)

Professor Portal

List Classes

SSN:

List Student Grades

Course Number:

Section Number:

[Back to Home](#)

Connected successfully

Course: calc - Classroom: 42 - Days: Wednesday - Time: 10am-12pm

Course: database - Classroom: 43 - Days: Thursday - Time: 10am-12pm

[Back to Home](#)

Query B:

Professor Portal

List Classes

SSN:

Find Classes

List Student Grades

Course Number:

Section Number:

Submit

[Back to Home](#)

Connected successfully

Grade (B): Count: 1

Grade (B-): Count: 2

[Back to Home](#)

Professor Portal

List Classes

SSN:

Find Classes

List Student Grades

Course Number:

332

Section Number:

1

Submit

[Back to Home](#)

Connected successfully

Grade (A): Count: 3

Grade (C+): Count: 1

Grade (C-): Count: 1

[Back to Home](#)

Student Input output

Query A:

Student Portal

Find Course Sections

Course Number:

Find Grades

Campus-Wide ID:

[Back to Home](#)

Connected successfully

Section Number: 2 - Classroom: 40 - Meeting Days: Monday - Time: 10am-12pm - Enrolled: 5

[Back to Home](#)

Student Portal

Find Course Sections

Course Number:

Find Grades

Campus-Wide ID:

[Back to Home](#)

Connected successfully

Section Number: 6 - Classroom: 44 - Meeting Days: Friday - Time: 10am-12pm - Enrolled: 3

[Back to Home](#)

Query B:

Student Portal

Find Course Sections

Course Number:

Find Grades

Campus-Wide ID:

[Back to Home](#)

Connected successfully

Courses Taken:

Course: database - Grade: A

Course: algo - Grade: A

Course: stats - Grade: C

[Back to Home](#)

Student Portal

Find Course Sections

Course Number:

Search

Find Grades

Campus-Wide ID:

Search

[Back to Home](#)

Connected successfully

Courses Taken:

Course: database - Grade: A

Course: algo - Grade: A

Course: stats - Grade: C

Course: calc - Grade: B-

[Back to Home](#)